

The Singularity Protocol:

A Deterministic Initialization and Stochastic Resource Framework for Competitive Commander (cEDH)

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Abstract

We present the **Singularity Protocol**, a proposed rule set amendment for the Competitive Commander (cEDH) format designed to optimize game initialization latency and mitigate the systemic First-Mover Advantage (FMA). The protocol replaces the stochastic London Mulligan with the **Event Horizon Draft**, a deterministic hand-construction algorithm that scales selection depth linearly against player seat index. Furthermore, we introduce **The Singularity**, a stochastic auxiliary resource mechanism that integrates the Vintage Restricted List into the Command Zone via a risk-gated access function. We demonstrate via hypergeometric analysis that this protocol increases the probability of Turn 0 interaction for the fourth seat by approximately 14.1% while reducing administrative downtime by an order of magnitude.

1 Introduction

The current operational state of cEDH is characterized by two primary inefficiencies: (1) excessive administrative overhead caused by recursive shuffling during the Mulligan phase, and (2) a statistically significant win-rate skew favoring the first player (Seat 1) due to the multiplayer nature of the format. Standard initialization protocols operate with time complexity $O(m \cdot t_{shuffle})$, where m is the average mulligan count.

The proposed **Singularity Protocol** reduces initialization to $O(1)$, creating a fixed-time setup phase. Additionally, by introducing a variable-depth drafting mechanic, we provide a mathematical counterbalance to the tempo advantage of the starting player, dampening the *Vanguard* vs. *Anchor* disparity.

2 Protocol Specification

2.1 Mechanism A: The Singularity (Σ)

To bridge the power disparity between cEDH and Vintage while introducing high-variance risk vectors, each player is allocated one auxiliary card slot.

Definition 2.1 (The Singularity Asset). Each player registers exactly one card, Σ , in their Command Zone. Σ must be a card currently listed on the **Vintage Restricted List** (e.g., *Black Lotus*, *Time Walk*, *Ancestral Recall*). Σ is public knowledge revealed prior to game initialization.

2.1.1 Constraints and Access

The asset Σ is subject to strict usage constraints to prevent format homogenization:

1. **Usage Cap:** Σ may be successfully cast exactly **once** per game instance. If Σ leaves the stack (via resolution or counterspell), it is exiled.
2. **Access Gating:** Σ cannot be cast via standard priority. It requires a successful *Breach Event*.

2.1.2 The Breach Protocol

Once per turn cycle, during their pre-combat main phase, the active player may attempt to access Σ by rolling a standard 6-sided **Planar Die** (D_p). Let the outcome space be $\Omega = \{\emptyset, \Phi, \mathcal{C}\}$.

- **Failure State** ($x = \emptyset$): The faces corresponding to blank results (4 faces).

$$P(\text{Fail}) = \frac{4}{6} \approx 66.7\% \quad (1)$$

Penalty: The player incurs **3 damage**. Σ remains locked in the Command Zone.

- **Success State** ($x \in \{\Phi, \mathcal{C}\}$): The faces corresponding to Planeswalker or Chaos symbols (2 faces).

$$P(\text{Success}) = \frac{2}{6} \approx 33.3\% \quad (2)$$

Effect: The player gains permission to cast Σ this turn, subject to standard mana costs and timing. *Planechase Clause:* If playing with Planechase, the Chaos symbol ($\mathcal{C}$) additionally triggers the active plane's chaos ability. In standard games, $\mathcal{C}$ is functionally identical to Φ .

2.2 Mechanism B: The Event Horizon Draft

To eliminate the latency of shuffling, the Mulligan phase is replaced by a deterministic selection algorithm.

Protocol 2.1 (Draft Algorithm). At game start ($t = 0$), each player i distributes cards from the top of their library face-down into 7 disjoint piles, denoted $P_{i,1}, \dots, P_{i,7}$. For each pile $j \in \{1, \dots, 7\}$:

1. **Selection:** The player views the contents of $P_{i,j}$ and selects exactly one card $c \in P_{i,j}$ to add to their hand.
2. **Tuck:** The remaining set $P_{i,j} \setminus \{c\}$ is placed on the bottom of the library.
3. **Integrity Broadcast:** Players must visually isolate the rejected cards to verify pile cardinality before bottoming them to ensure no card "pocketing" occurs.

2.2.1 Turn Order Compensation (TOC)

To counteract FMA, the cardinality of the piles $|P_{i,j}|$ scales with the seat index i . The *Selection Depth* D_i is defined as the total number of cards viewed. The configurations are optimized to grant later seats deeper access to their library.

Table 1: Selection Depth as a Function of Seat Index

Seat (i)	Role	Pile Configuration	Total Depth (D_i)	Selection Ratio
1	Vanguard	$7 \times [2]$	14	50.0%
2	—	$5 \times [2], 2 \times [3]$	16	43.8%
3	—	$3 \times [2], 4 \times [3]$	18	38.9%
4	Anchor	$1 \times [2], 6 \times [3]$	20	35.0%

3 Mathematical Analysis

3.1 Hypergeometric Probability of Early Interaction

A critical metric for format health is the ability of Seat 4 (The Anchor) to interact with a Turn 1 win attempt by Seat 1. Let $K = 6$ be the number of *free* interaction spells in a deck (e.g., *Force of Will*, *Pact of Negation*) and library size $N = 99$.

The probability of finding at least one interactive spell in the opening draft is given by the cumulative hypergeometric distribution:

$$P(X \geq 1) = 1 - \frac{\binom{N-K}{D_i}}{\binom{N}{D_i}} \quad (3)$$

Comparing Seat 1 ($D_1 = 14$) vs Seat 4 ($D_4 = 20$):

$$P(X_1 \geq 1) = 1 - \frac{\binom{93}{14}}{\binom{99}{14}} \approx 0.605 \quad (4)$$

$$P(X_4 \geq 1) = 1 - \frac{\binom{93}{20}}{\binom{99}{20}} \approx 0.746 \quad (5)$$

Proposition 3.1. *The protocol provides Seat 4 with a **+14.1% statistical advantage** in finding defensive tools compared to Seat 1. This differential acts as a dampening coefficient on the inherent First-Mover Advantage.*

3.2 Risk Assessment of The Singularity

The Breach mechanic follows a geometric distribution with success probability $p = 1/3$. The expected number of turns $\mathbb{E}[T]$ to access the Singularity is:

$$\mathbb{E}[T] = \frac{1}{p} = 3 \text{ turns} \quad (6)$$

The expected life loss $\mathbb{E}[L]$ is a function of the expected number of failures ($\mathbb{E}[T] - 1$) and the damage penalty ($\delta = 3$):

$$\mathbb{E}[L] = (\mathbb{E}[T] - 1) \times \delta = (3 - 1) \times 3 = 6 \text{ life} \quad (7)$$

This cost structure ensures that while the power ceiling of the format is raised, the *expected utility* is regulated by a significant tempo (3 turns) and resource (6 life) tax. This prevents the mechanic from strictly homogenizing the early game in favor of aggressive *Ad Nauseam* strategies.

4 Conclusion

The **Singularity Protocol** offers a rigorous solution to the logistical and game-theoretic imbalances of the London Mulligan in multiplayer environments. By shifting initialization from stochastic permutation (shuffling) to deterministic selection (drafting), and by scaling information access against turn order, the protocol optimizes for both time efficiency and competitive equity.